

POLS 15 – Section 11

Course Review

Concepts

Causality & Correlation

- Is this just a matter of experimental studies versus observational studies? Why or why not?
- Why can't we infer causality from observational studies?
 - Selection problems
 - Confounding
 - Comparability

Variables

- Independent Variable (IV): a variable that possibly influences the value of the dependent variable. It is usually denoted as X .
- Dependent Variable (DV): the outcome of interest, usually denoted as Y .
- In an experimental study, the IV can also be called the “treatment” and is usually denoted as D .

Cycle of Statistical Inference

- Estimand $\rightarrow$ Sample $\rightarrow$ Estimator $\rightarrow$ Estimate
- Other terms for estimand: population, true value, parameter
- Examples of estimators: sample mean, sample variance, difference-in-means, OLS regression
- Examples of estimates: ATE, β , $\bar{X}$

Validity

- Internal Validity: A finding is internally valid when it is based on a process that is free from systematic error.
 - i.e., confounders are properly controlled
- External Validity: A finding is externally valid when it applies beyond the context in which the analysis was conducted (generalizability).
- Experimental results are often considered internally valid but their external validity can be debated.

Endogeneity & Exogeneity

- An independent variable is endogenous if changes in it are related to other factors that influence the dependent variable.
 - i.e., $Cor(X, \epsilon) \neq 0$
- An independent variable is exogenous if changes in it are unrelated to other factors that influence the dependent variable.
 - i.e., $Cor(X, \epsilon) = 0$

Random Variables

- Three characteristics:
 - There is a set of possible values
 - Each value occurs with some probability
 - The value is not known until there is a draw (i.e., the event happens)
- Probabilities aggregate to 1 (100%)

P-Values & Hypothesis Testing

- P-Value: the probability of observing a coefficient as high as we actually observe if the null hypothesis is true
 - $H_0 : \beta = 0$
 - $H_A : \beta \neq 0$
- What is the significance level α ?
- Interpretation:
 - Statistically significant or insignificant
 - Reject or fail to reject the null hypothesis
 - NEVER ACCEPT

Confidence Intervals

- Confidence intervals define the range of true values that are consistent with the observed coefficient estimate.
 - Depend on the point estimate (coefficient) and measure of uncertainty.

$$CI = \hat{\beta} \pm t^* * SE_{\hat{\beta}}$$

- t^* changes depending on confidence level.
 - At 95% confidence ($\alpha = 0.05$), $t^* = \pm 1.96$

Research Design

Experiments

- Main characteristic: randomly assign participants to control or treatment groups
 - Randomness $\neq$ Randomization
- ABC Issues:
 - Attrition: occurs when people drop out of an experiment altogether so that we do not observe the DV for them.
 - Balance: groups are balanced if the distributions of control variables are the same for the treatment and control groups (no statistical difference).
 - Compliance: a compliance problem occurs when subjects assigned to the treatment group do not actually receive the treatment, often because they opt out in some way.

Surveys

- Figure out your research question and establish your target population.
- Key considerations:
 - How to sample your population (random? purposive? quota?)
 - Survey instrument & delivery (questionnaire, phone call, in-person, etc.)
- Types of survey and administrative data:
 - Cross-sectional data: data having observations for multiple units for one time period.
 - Time-series/longitudinal data: data having observations over time.
 - Panel data: data having observations for the same individuals at multiple points in time.

Qualitative Research

- Case selection strategies affect what types of questions you can answer.
- Types of cases:
 - Typical
 - Diverse
 - Extreme
 - Deviant
 - Influential
 - Most similar
 - Most different

Analysis

OLS Models

- When do we use bivariate versus multivariate models?
 - Why do we need more IVs?
 - In an OLS model, what does each term represent?
- Points to review:
 - Coefficient interpretation (intercept & slope)
 - Goodness of fit (R^2)
 - Model specification
 - Null/alternative hypotheses & p-values
 - Number of IVs
 - Robustness
 - Model fishing

Research Papers

How to Read a Paper

- Key points:
 - Research topic
 - IVs & DVs + measurement
 - Research design & methods
 - Findings & implications
 - Thoughts on improving the research (different measurements? methods? design?)

Reading List

- Wand et al. (2001), “The Butterfly Did It: The Aberrant Vote for Buchanan in Palm Beach County, Florida”
- Kalla & Broockman (2015), “Congressional Officials Grant Access Due to Campaign Contributions: A Randomized Field Experiment”
- Colantone et al. (2024), “The Political Consequences of Green Policies: Evidence from Italy”
- Gerber, Green, & Larimer (2008), “Social Pressure & Voter Turnout: Evidence from a Large-Scale Field Experiment”

Reading List

- Gerring (2007), “The Case Study: What It Is and What It Does”
- Seawright & Gerring (2008), “Case Selection Techniques in Case Study Research: A Menu of Qualitative and Quantitative Options”
- Imai (2018), *Quantitative Social Science: An Introduction*, Ch. 5